

# Microbiology Lessons





## About the Course

An elective in Life Science, Microbiology is a challenging course for the Alveary senior. Some self-direction is required.

This topic is included in the following course(s): Science: Microbiology



## Placement & Combining Tips

Students should take General Biology before Microbiology. Note that there is some sensitive content in this course, and it is recommended for students in Grade 12.



## Scheduling

| GRADE | SCHEDULE INFO.                              | BOOKS                                                                                                                                                                                                     |
|-------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12    | Microbiology Lessons<br>5 times/week 45 min | A Field Guide to Bacteria<br>A Planet of Viruses<br>Growing a Revolution: Bringing Our Soil Back to Life<br>I Contain Multitudes<br>Micro Life<br>Microbe Hunters<br>The Immortal Life of Henrietta Lacks |

### Sample Weekly View

| Day 1                                                        | Day 2                | Day 3                | Day 4                                                | Day 5                                     |
|--------------------------------------------------------------|----------------------|----------------------|------------------------------------------------------|-------------------------------------------|
| <b>Science: Microbiology</b>                                 |                      |                      |                                                      |                                           |
| Microbiology Lessons<br>Nature Walks & Scouting: Grades 9-12 | Microbiology Lessons | Microbiology Lessons | Microbiology Lessons<br>Nature Notebook: Grades 9-12 | Microbiology Lessons<br>Microbiology Labs |



## Planning & Prep

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**LINKS:** Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

### Microbiology Lessons

- ☐ Select a science book from the Alveary bookshelf for personal reading time, as appropriate.
- ☐ Bookmark or print Quick Links, as needed.
- ☐ Gather supplies for the first microbiology labs initially. You will gather additional supplies once you decide on additional labs.

□ Teacher Note: Preread Chapter 15. Henrietta Lacks's children had to deal with a lot of abuse of every kind. Discuss with your student(s). The note for the student is in Week 1 of Term 2.



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

### Microbiology Lessons



A Field Guide to Bacteria



A Planet of Viruses



Growing a Revolution: Bringing Our Soil Back to Life



I Contain Multitudes



Micro Life



Microbe Hunters



The Immortal Life of Henrietta Lacks



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



## Quick Links

### Microbiology Lessons

- ∞ [Nature Journal Connection](#)
- ∞ [Alveary Bookshelf](#)
- ∞ [Illustrated Guide to Home Biology Experiments](#)
- ∞ [Seven Wonders of the Microbe World](#)
- ∞ [Microbiology from A to Z](#)
- ∞ [Microbiology Current Events](#)

Click THIS text  
or scan the QR  
code for links.



# Microbiology Lessons

## How To Approach



### Introduce

- If starting a new book or a new topic in the book, then look at the title or a picture and take a moment to consider previous ideas and experiences.
- If continuing a previous reading, recap what was read previously. Often, the title of the book's section can help to draw out the main idea.



### Read

- Read or do, as instructed in the lessons, making note of or flagging unfamiliar terms, interesting ideas, important dates, inspiring quotes, etc.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- As they read, learners record ideas in a notebook or binder using outlines, diagrams, graphic organizers, or other methods (or a combination of methods) that suit them. These recordings can be a helpful mechanism for remembering or a mini-narration to support understanding.
- If learners do not understand a word or concept, do not worry. Try reading over the passage again, studying a picture or diagram, connecting the idea to something from real life, or practicing chapter exercises. The lab/field work further supports major concepts from the text.



### Narrate

- Process the ideas of the lesson by retelling, defining a concept, explaining the links in a chain of thought, etc. Do this orally or silently to yourself.
- Use words, pictures, outlines, etc.
- If a particular idea cannot be narrated, then read or examine the text again.



### Discuss

- Consider with the teacher any thoughts, confusion, or concerns about the passage.
- If understanding is still uncertain, try rereading the passage or do some personal research on the topic.



### Connect

- Follow any extra links, examine any sidebars in the text, or pursue additional reading, depending on student interest.

# Microbiology Lessons

[Click THIS text or scan the QR code for links.](#)



## Term 1

### WEEK 1 45m Microbiology Lessons - Lesson 1

Materials: I Contain Multitudes

#### → READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

#### • PLAN WEEKLY

- nature walk - record observations
- science free read

### WEEK 1 45m Microbiology Lessons - Lesson 2

Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week, starting with the Appendices. Continue with the Introduction and read forward. Sometimes there will be practical things to do with the microscope and bacteria, but other times you will be reading to gain a greater understanding of bacteria from many different places. Sometime before next week, watch the author's video on bacteria. You may or may not agree with the author's perspective, but the point of the video is to see where some of the more recent thinking is concerning microbes.

#### → VIEW

∞ Video Link: Talk by Betsey Dexter

### WEEK 1 45m Microbiology Lessons - Lesson 3

Materials: Microbe Hunters or A Planet of Viruses

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 1 45m Microbiology Lessons - Lesson 4

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

#### ★ STUDENT/TEACHER TIP

Gather some family and friends and watch the video "Seven Wonders of the Microbe World."

∞ Video Link: Seven Wonders

### WEEK 1 45m Microbiology Lessons - Lesson 5

Materials: Illustrated Guide to Home Biology Experiments

#### → READ

Read p.3-51 at your own pace over the next few lab periods. If you already have a good working microscope of reasonable quality, you may be able to skim through the microscope chapter. If you are not that familiar with your microscope, you may benefit from reading it. You will move at your own pace through the microbiology labs in this book after completing the first 6 in order (Labs I-1 through II-3). Then, you are free to pick and choose based on your interests and/or the availability of needed supplies. The more you do, the more you will grow in your understanding of microbiology! There are several labs in the book that are general biology labs (10-12,18,20,21,23, and 24). You will skip over these and focus on the ones that require a microscope.

### WEEK 2 45m Microbiology Lessons - Lesson 6

Materials: I Contain Multitudes

#### → READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

#### • PLAN WEEKLY

- nature walk - record observations
- science free read

# Microbiology Lessons

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## Term 1

### WEEK 2 45m Microbiology Lessons - Lesson 7

Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Come up with an interesting way to present or summarize the information in this book. It could be a map, chart, diagram, or something else.

### WEEK 2 45m Microbiology Lessons - Lesson 8

Materials: Microbe Hunters or A Planet of Viruses

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 2 45m Microbiology Lessons - Lesson 9

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 2 45m Microbiology Lessons - Lesson 10

Materials: Illustrated Guide to Home Biology Experiments

#### → READ

Finish reading p.3-51.

### WEEK 3 45m Microbiology Lessons - Lesson 11

Materials: I Contain Multitudes

#### → READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 3 45m Microbiology Lessons - Lesson 12

Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 3 45m Microbiology Lessons - Lesson 13

Materials: Microbe Hunters or A Planet of Viruses

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 3 45m Microbiology Lessons - Lesson 14

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 3 45m Microbiology Lessons - Lesson 15

Materials: Illustrated Guide to Home Biology Experiments

# Microbiology Lessons

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Term 1

→ LAB DAY

SAMPLE